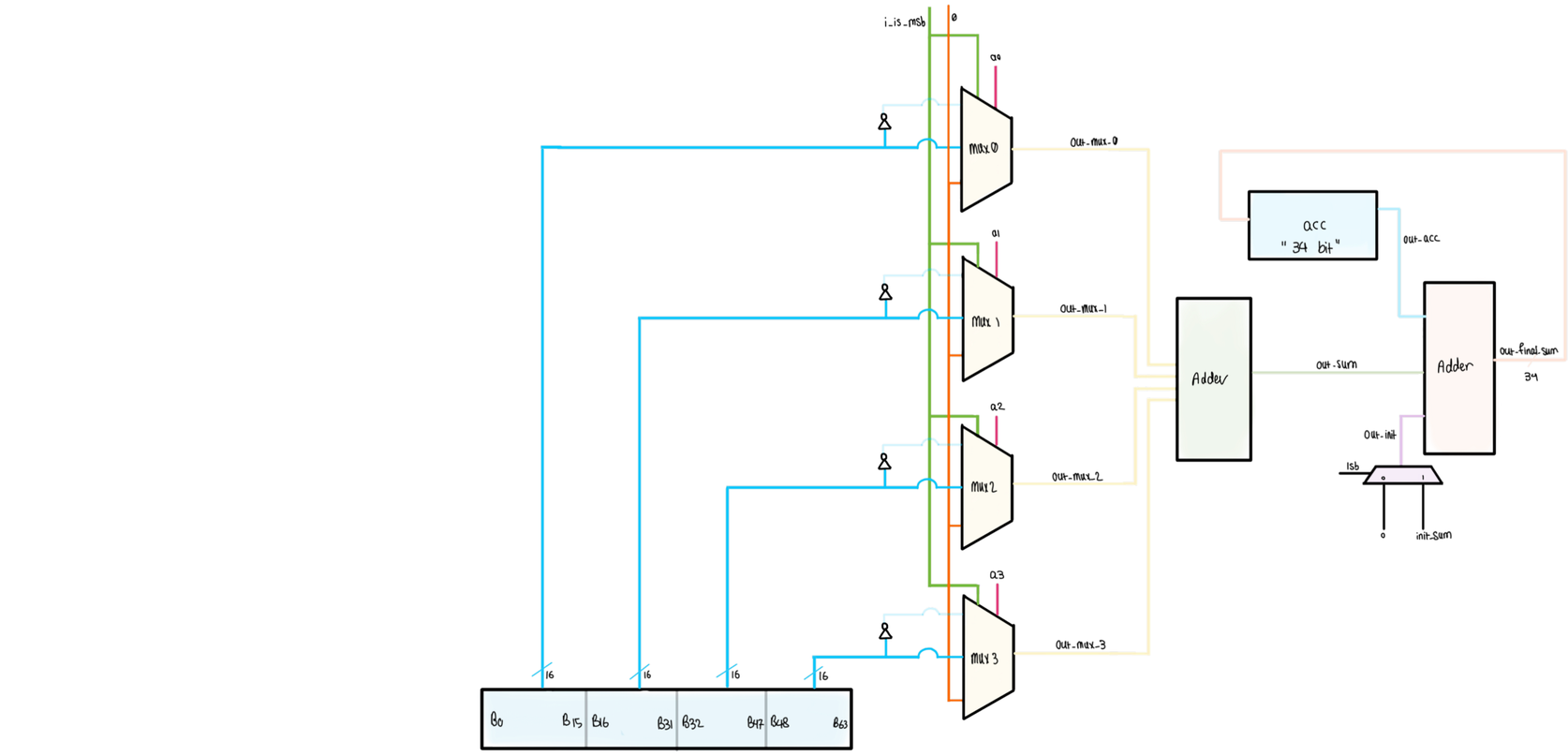
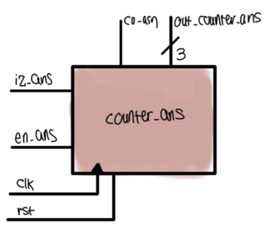
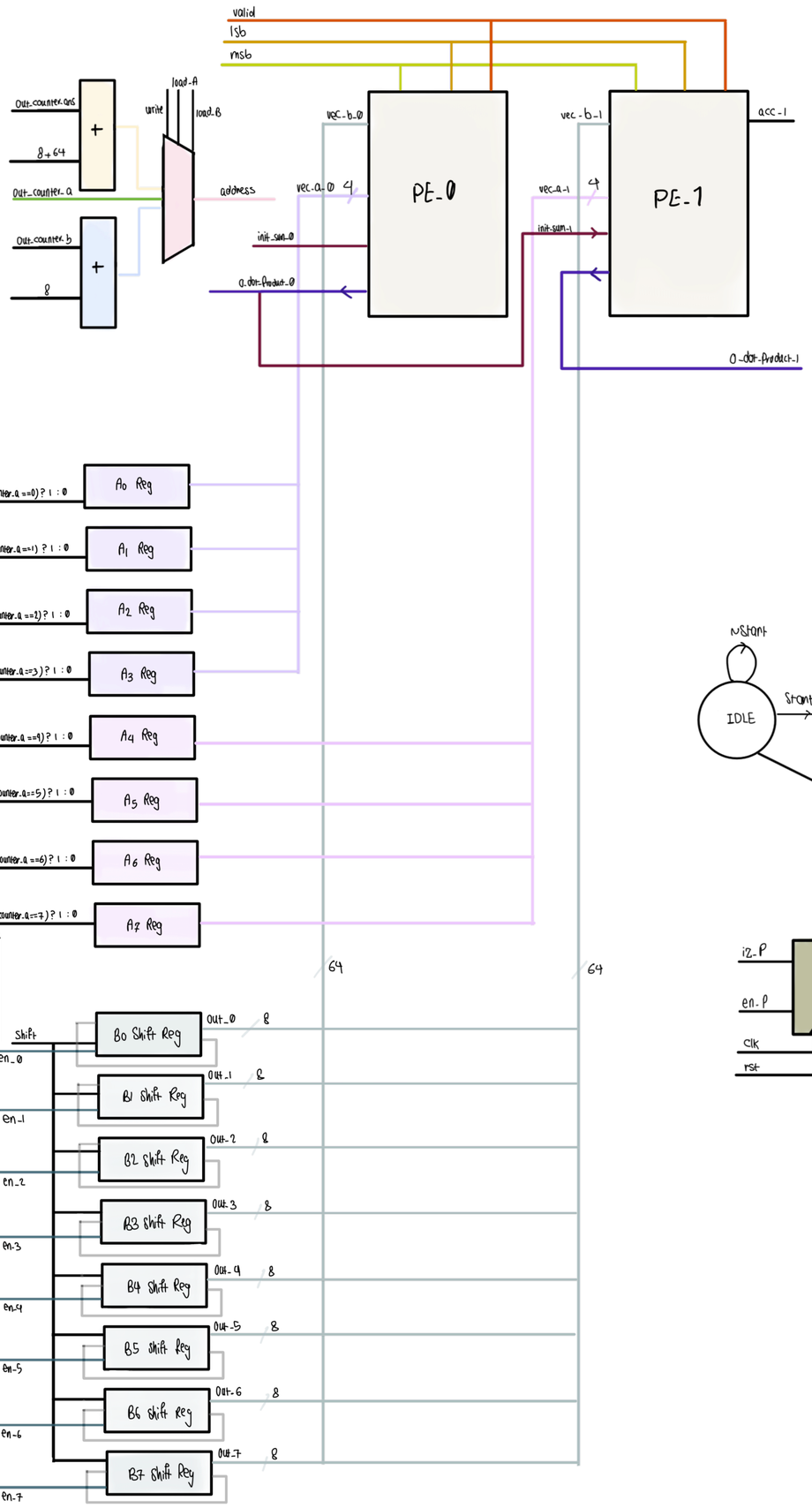
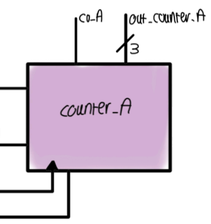
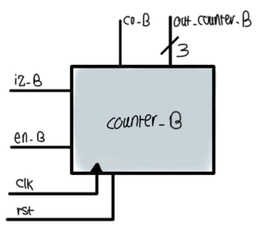

i_is_msb
0
a0
Mux 0
out_mux_0
a1
Mux 1
out_mux_1
a2
Mux 2
out_mux_2
a3
Mux 3
out_mux_3
16
16
16
16
B0
B15
B16
B31
B32
B47
B48
B63
Adder
out_sum
acc
" 34 bit "
out_acc
Adder
out_final_sum
34
out_init
lsb
0
1
0
init_sum

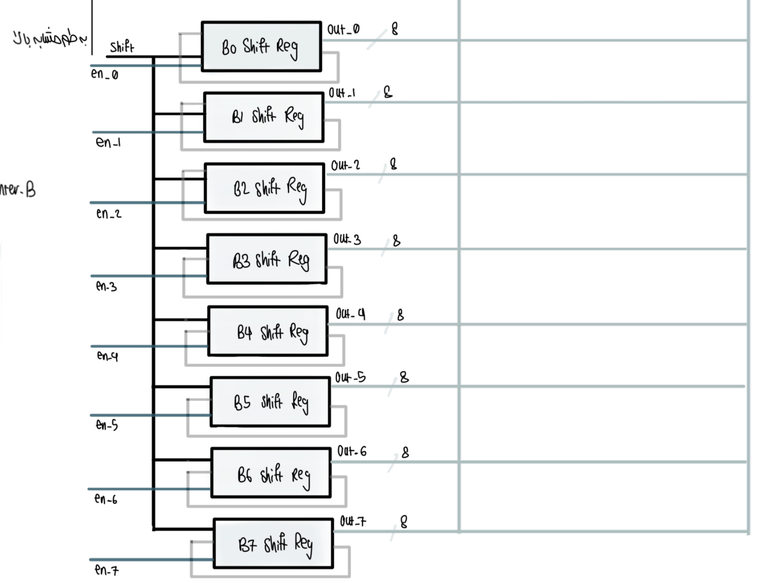

co_ans
out_counter_ans
3
counter_ans
i2_ans
en_ans
clk
rst

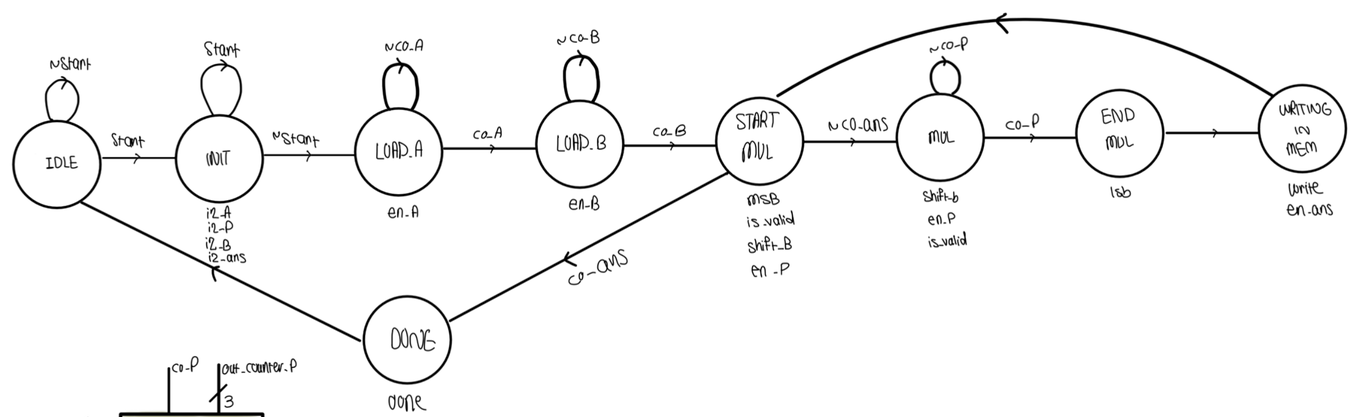

valid
lsb
msb
out_counter_ans
8'd64
+
load_A
write
load_B
out_counter_a
address
out_counter_b
8
+
vec_b_0
vec_a_0
4
PE_0
init_sum_0
a_dot_product_0
vec_b_1
vec_a_1
4
PE_1
acc_1
b_dot_product_1

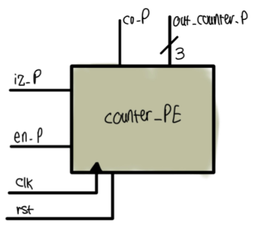

co_A
out_counter_A
3
counter_A
i2_A
en_A
clk
rst


co_B
out_counter_B
3
counter_B
i2_B
en_B
clk
rst

(out_counter_a ==0) ? 1 : 0
A0 Reg
(out_counter_a ==1) ? 1 : 0
A1 Reg
(out_counter_a ==2) ? 1 : 0
A2 Reg
(out_counter_a ==3) ? 1 : 0
A3 Reg
(out_counter_a ==4) ? 1 : 0
A4 Reg
(out_counter_a ==5) ? 1 : 0
A5 Reg
(out_counter_a ==6) ? 1 : 0
A6 Reg
(out_counter_a ==7) ? 1 : 0
A7 Reg


64
64
Shift
en_0
B0 Shift Reg
out_0
8
en_1
B1 Shift Reg
out_1
8
en_2
B2 Shift Reg
out_2
8
en_3
B3 Shift Reg
out_3
8
en_4
B4 Shift Reg
out_4
8
en_5
B5 shift Reg
out_5
8
en_6
B6 shift Reg
out_6
8
en_7
B7 Shift Reg
out_7
8


Nstart
IDLE
Start
INIT
i2_A
i2_P
i2_B
i2_ans
Nstart
LOAD_A
en_A
Nco_A
ca_A
LOAD_B
en_B
Nca_B
ca_B
START MUL
msb
is_valid
shift_B
en_P
N co_ans
MUL
shift_b
en_P
is_valid
Nco_P
co_P
END MUL
lsb
WRITING IN MEM
write
en_ans
co_ans
DONE
done


co_P
out_counter_P
3
counter_PE
i2_P
en_P
clk
rst